

Jay Rao

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EDUCATION

Northwestern University, Evanston, IL

Expected December 2027

Bachelor of Arts (B.A.) in Computer Science, Concentration in Artificial Intelligence

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Scalable Software Architecture, Discrete Math, Fundamentals of Programming I, 1.5 & II, Software Studio

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, TypeScript, SQL

Frameworks & Libraries: FastAPI, Node.js/Express, React, PyTorch, NumPy, Pandas, Playwright, pytest

Tools & Platforms: Git/GitHub, GitHub Actions (CI/CD), AWS (EC2, ECS, RDS, S3, SES, IAM, Secrets Manager), Terraform, Docker, PostgreSQL, MySQL, Linux/Bash, Agile/Scrum

EXPERIENCE

Founding Engineer | GridGate LLC

Feb. 2026 – Present

- Sole engineer on GridGate Queue Analysis v1.0, an end to end data platform built around a 19 source data lake fed by 20+ scrapers, a 9 module feature pipeline, and a modeling layer covering 3,700+ MISO interconnection projects
- Automated ingestion of 300,000+ state and county regulatory filings and meeting minute PDFs across 15 states and 1,000+ counties using fault tolerant scrapers with retry logic plus LLM driven discovery for non standard sites
- Trained and validated two CatBoost classifiers covering project success and withdrawal (0.93 ROC-AUC, 85% accuracy) and 5 class withdrawal phase (0.94 one vs rest ROC-AUC), each with SHAP explainability for stakeholders
- Designed a 6 tier entity resolution pipeline mapping 3,700+ projects to parent developers at 83% precision via document grounded lookups and multi stage evidence verification against public filings
- Built two separate sentiment pipelines, one scraping social media (Facebook, X, Reddit, YouTube) and another crawling local newspaper archives across the MISO footprint, to surface early community opposition signals ahead of filing

Technology Manager | iloveyou LLC

Feb. 2026 – Present

- Lead developer managing 3 junior engineers across sprint planning, task assignment and code review, taking a consumer social app from an empty repository to a shipped v1.0 production release
- Built the backend from scratch in Node.js/Express on PostgreSQL with JWT auth and refresh token rotation, email and phone verification (Amazon SES, Twilio Verify), APNs push notifications and REST APIs, applying secure coding practices for credential storage and personally identifiable user data across every endpoint the mobile client calls
- Implemented the full iOS client in SwiftUI (MVVM) covering onboarding, community assignment, profile creation, feed, messaging, events, and groups from the very first wireframe through to production release
- Engineered AWS infrastructure as code in Terraform (VPC, ECS Fargate, ALB, RDS, S3, CloudFront, SES, Secrets Manager, IAM) with remote state and DynamoDB locking, plus CI/CD pipelines in GitHub Actions

Computer Science Educator | Northwestern TIDAL & tilt Labs

Jan. 2026 – Jun. 2026

- Co-led two fifth grade classrooms in Evanston/Skokie School District D65 each week, guiding 40+ students through a project based Python curriculum built on TunePad and SportSense tools

PROJECTS

game.sense | Python, PyTorch, PyTorch Geometric, FastAPI, PyQt6, pytest

- Building a spatial intelligence engine that predicts ball and vehicle positions up to 5 seconds ahead from a single occluded viewpoint, chaining a 4 layer GATv2 graph encoder into per entity LSTMs and a Gaussian mixture decoder trained on 2,193 parsed replays
- Cut 1 second ball position error from an 11.39m baseline to 3.06m over five retraining rounds, sharing one frustum and occlusion masking core between offline label generation and live inference so training and serving geometry can never silently drift apart
- Shipped the live path as a Kalman filter for visible entities plus frame budgeted model inference behind a PyQt6 overlay and a FastAPI WebSocket server, validated by a 277 test pytest suite spanning separate CUDA and CPU training/inference environments

nuPython Interpreter & Debugger | C, C++, STL, Valgrind

- Built a complete interpreter for a Python like language in 2,000+ lines of C/C++, including a recursive expression evaluator with automatic int/float coercion, control flow, built in functions and runtime type checking across integers, floats, strings, booleans, pointers and None, each with descriptive semantic error messages surfaced back to the caller
- Designed a simulated RAM pairing a dynamically resizing cell array with an alphabetically sorted name to address map, giving $O(\log n)$ binary search lookup with deep copying and leak free teardown verified under Valgrind on every run
- Implemented a GDB style source level debugger that unlinks and relinks the program graph to single step execution, supporting live variable inspection, memory dumps and full execution history tracking

Landscape Guessr | Ruby on Rails, PostgreSQL, PostGIS, JavaScript, Stimulus, Tailwind CSS

- Co-developed a GeoGuessr style web game in a 4 person Agile team on Rails 8 and PostgreSQL, deployed live to Heroku with S3 backed uploads and a seeded catalog of 1,400+ real world Wikidata landmarks
- Built a server authoritative real time multiplayer mode with polling backed lobbies, deadline enforced rounds and idempotent round ending logic that never trusts client clocks, plus a live global leaderboard
- Modeled a region hierarchy with PostGIS backed filtered image sets, and added an AI assisted set builder that turns a natural language prompt into a Wikidata query via Gemini function calling and preview

vibes.fm | React, Node.js, Express, MySQL, AWS, Transformers.js, OAuth 2.0

- Engineered a full stack playlist generator that embeds a natural language description of every track (Spotify metadata, audio features, genres, lyrics) into a 384 dimensional vector with a local sentence transformer, then ranks an in memory cache by cosine similarity against the user prompt and exports matches to Spotify via PKCE OAuth 2.0, all deployed on EC2 behind nginx and PM2